26-382-0106- DATA STRUCTURES LAB

1. Write a C program to evaluate the arithmetic expression ((a -b / c * d + e) * (f +g)) and display its solution.
2. Create a C Program to read 3 integer values, find the largest among them.
3. Develop a C program to check if the number is Prime or not.
4. Develop a program to check whether the given number is armstrong or not.
5. Write a c program to read a string and count the number of vowels, consonants and spaces in it.
6. Using structure, write a program to read and print data of n employees (Name, Employee Id and Salary)
7. Write a C program to create a fibonacci series using recursive function.
8. Find the factorial of a given Natural Number n using
    a. i) a non recursive function
    b. ii) a recursive function
9. Do the following using pointers
    a. i) add two numbers
    b. ii) swap two numbers using user defined function
10. Construct binary search trees to perform insertion, deletion, search
11. Apply Queue and stack in Breadth First Search and Depth First Search respectively
12. Write a program to create a binary search tree and find the number of leaf nodes
13. Create a binary search tree with the following operations:
    a. i)Insert a new node .
    b. ii)Inorder traversal.
    c. iii)Preorder traversal.
    d. iv)Postorder traversal.
    e. v)Delete a node.
14. Design, develop, and execute a program in C to create a max heap and perform the following operations
    a. i)Insertion
    b. ii)Deletion
    c. iii)Print Largest Value
15. Write a program to implement Depth First Search (DFS) graph traversal methods.
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17. Create a text file containing the name, height, weight of the students in a class. Perform Quick sort and Merge sort on this data and store the resultant data in two separate files. Also write the time taken by the two sorting methods into the respective files.
    i. Sony Mathew 5.5 60
    ii. Arun Sajeev 5.7 58
    iii. Rajesh Kumar 6.1 70
18. Write a program to sort a set of numbers using Heap sort and find a particular number from the sorted set using Binary Search.
19. Implement a Hash table using the Chaining method. Let the size of hash table be 10 so that the index varies from 0 to 9.
20. Implement a Hash table that uses Linear Probing for collision resolution